DATE:	**TRANSPORTATION PROBLEM**
EXP.NO: 2	

AIM:

To solve the Transportation Problem for minimizing the cost of transporting goods from multiple sources to multiple destinations using Python.

PROCEDURE:

1. **Install Required Libraries:**

o Make sure you have numpy and scipy installed. You can install them using pip if not already installed.

```
pip install numpy scipy
```

2. **Define the Problem:**

o Identify the cost matrix, supply array, and demand array.

3. **Set Up the Problem in Python:**

o Use the scipy.optimize.linprog function to set up and solve the Transportation Problem.

4. **Run the Code:**

o Execute the Python code to get the optimized result.

5. **Analyze the Output:**

Interpret the results and validate the solution against the problem constraints.

PROBLEM 1:

Source	D1	D2	Supply
S1	2	4	20
S2	5	3	30
Demand	20	30	

CODE:

```
import numpy as np
from scipy.optimize import linprog

# Cost
coefficients
cost = [2, 4, 5,
3]

# Equality constraints (Supply constraints) A_eq = [
        [1, 1, 0, 0],
        [0, 0, 1, 1]
]

b_eq = [20, 30]

# Solve
result = linprog(cost, A_eq=A_eq, b_eq=b_eq)

# Output
print("Minimum Transportation Cost =", result.fun)
print("Optimal Solution =", result.x)
```

Output:

```
Minimum Transportation Cost = 150.0
Optimal Solution = [20. 0. 0. 30.]
```

RESULT:

Thus the Transportation Problem for minimizing the cost of transporting goods from multiple sources to multiple destinations using Python was executed successfully.